

Lab 7: Base R Chart Selection

Level: Intermediate Mapping: LO4 | K3 | A3 | A4 Concepts: scatter, bar, histogram, labels, interpretation

Scenario

A transport team needs three views of fuel-economy and transmission data.

Goal

A scatter plot, bar chart and histogram saved as PNG files.

Files in this folder

- lab-07.R - completed guided R script
- verify.R - repeatable acceptance check
- lab-workbook.xlsx - brief, input data, controls and evidence record
- input-data.csv - portable fallback if readxl is unavailable

Detailed step-by-step

1. Import the workbook's vehicle data.
2. State one question for each required chart.
3. Create a labelled scatter plot of weight versus fuel economy.
4. Create a zero-based bar chart of transmission counts and a histogram of fuel economy.
5. Save the plots and write one evidence-based interpretation per chart.

Run and verify

1. Set the RStudio working directory to this lab folder.
2. Open lab-07.R and read the header, inputs and validation rules.
3. Run the complete script with `source("lab-07.R")`.
4. Confirm the Console ends with LAB VERIFIED: outputs created and checks passed.
5. Run `source("verify.R")` and retain the verification message.
6. Open analysis_output.csv and compare the evidence with the workbook Control sheet.
7. Complete the workbook Evidence sheet and save it without changing the filename.

Acceptance criteria

- [] Each chart answers a distinct question.
- [] Axes, units and categories are readable.
- [] Interpretations describe evidence without claiming causation.

Troubleshooting

- there is no package called 'readxl': the script automatically uses input-data.csv; install readxl later if Excel import is required.

- cannot open file: confirm RStudio's working directory is this exact lab folder.
- object not found: run the script from the top so imports and validations occur before analysis.
- Verification fails: inspect the first error, compare column names with Input_Data, and rerun from a clean session.

Evidence to submit

Completed Excel Evidence sheet, R script, generated output files and the final Console verification message.